



OpenKM – Low-Data Translator for Transport



The gap in Cape Town mobility

Informal feeders lose 25% downtime to punctures, no data, no safety net. City needs integration without owning the fleet.

- Mountainside 15% grades** kill cheap e-bikes — motor overheat, battery stranded at 60%
- Punctures = rider stuck** on Mountain Road, no roadside repair, taxi rank missed
- MyCiTi feeder gap 0.8-2km** uncovered, taxi queue risky after dark
- No open km standard** to calculate cost per km or side-road patch savings

25%

downtime today

0

open km audited

15%

grade barrier

Solution is a **translator**, not another fleet

Support not own — Gordon's Bay pilot at Checkers hubs. One open km standard that translates informal, e-bike, MyCiTi, taxi and fuel costs to patch side roads and fund safe feeder hubs.

open km = base × method × fuel index

Low-data SQLite + GPS. Works offline, syncs at hub WiFi. Any network translated to same cost per km. City pays per feeder km, not per bike.



Low-data SQLite core

2-sec GPS ping, offline cache, 14kb per trip. No signal needed at rank. Anonymous ID, 100m rounding for privacy.

Offline-first

IP65 sealed



Checkers hub share

Solar LED, CCTV, security, roof already paid. DropBox bolted to wall, battery cabinet metal ventilated. No new build.

5 sites ready

Shared cost



Support not own jobs

Time Controller = TapCard agent + maintenance + safety trainer. Taxi rank partnership not competition. 3 jobs per hub.

Micro-enterprise

15-min onboard

Fleet dual — built for Gordon's Bay

HillClimber for Mountainside, CiTiDriver for flat MyCiTi feeders. Same OS, same odometer, same safety.

HillClimber Mountainside

15% GRADE

- 26" 85Nm mid-drive, 625Wh + spare, motor derate 80°C
- 180mm hydraulic disc, low-gear both-brakes weight-back protocol
- Tannus 100% airless puncture proof, 5,000km rated
- Hall sealed IP65 + OpenKM GPS dual odometer
- Wet Grip StrapOn compatible — 60-sec V-tread

10 units pilot Swap at hub

CiTiDriver Feeder

0.8-2KM FLAT

- 700c 65Nm, 625Wh, efficient on flat MyCiTi
- 160mm hydraulic, weekly QR log, brake cut-off
- Tannus airless — 0 roadside repairs, 300 days
- Same dual odometer, same ESP32-S3 AI brain
- CiTi sleeve Wet Grip — for winter slick

10 units pilot 25 km/h cap legal

Common: Helmet + high-vis + 400 lumen always-on • Spoke reflectors • Tilt sensor • 2x security shared Checkers

Both NFC PN532 0.2s

Safety is infrastructure

Safety measurements enable scale — City approval requirement. No open km released without training flag and QR log.



Rider

15-min

- Helmet, high-vis, 400 lumen always-on
- 25 km/h cap SA legal, training flag required
- No unlock on 15% grade without Hill pass



Bike

IP65

- 180/160mm hydraulic, weekly QR log
- Brake cut-off + tilt sensor, spoke reflectors
- Tannus airless — no roadside repair needed



Hill

Auto-cut

- Low gear, both brakes, weight back protocol
- 20% speed cut in rain via weather API
- Motor derate at 80°C, walk assist above 15%



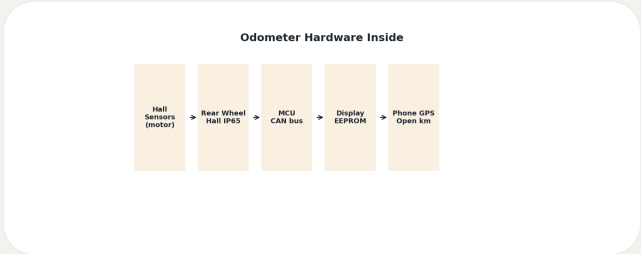
Hub

Checkers

- 2× security shared Checkers, CCTV solar
- Battery cabinet metal ventilated + smoke + extinguisher
- First aid, emergency button, offline binder

Odometer + AI chip inside display

Hardware km for service — sealed IP65 for False Bay salt. Open km for business — the translator City can audit. One display, two ledgers.



01

Hall sensors inside motor

3x Hall in mid-drive + rear wheel Hall backup. No spoke magnet to lose. Sealed IP65 against salt spray.

Pulse source • 1 rotation = 1 count

02

MCU CAN bus → EEPROM

Controller converts pulses to km/h using wheel size (26" vs 700c). Total km stored in EEPROM, survives battery removal.

Service ledger

03

Display head + GPS open km

Hardware odo on screen + phone GPS SQLite logs open km offline. Drift check flags maintenance.

Audit = HW vs GPS

04

ESP32-S3 0.5W TinyML

Inside display: BLE + PN532 NFC same board. Runs 4 brains offline, 0.5W from 36V buck.

Phase1 phone NPU, Phase2 open display

Hill brain

Torque + gradient, derate at 80°C, low gear prompt

Wet slip brain

Wheel vs GPS slip detection → Wet Kit + 20% cap

Maintenance brain

Vibration + odo predicts pads, Tannus 5k km, chain

Theft brain

Accel movement when locked → buzz hub security

Wet grip loses, StrapOn fixes it. NFC makes it offline

Airless Tannus is puncture-proof but loses wet grip — Mountainside rain needs extra bite. Hub issues kit, NFC logs it.



Wet Kit — extra grip strap-on

Nylon straps + cleated nails with 45a durometer glue-like rubber over 60a base, double-sided V-tread grips tyre + ground. 60-sec clip-on for 26" HillClimber, sleeve for 700c CiTiDriver. Stored at hub, issued when rain >70% via weather API.

- reusable
- roll without damage
- 20% speed cut auto



NFC edge-to-edge cycle

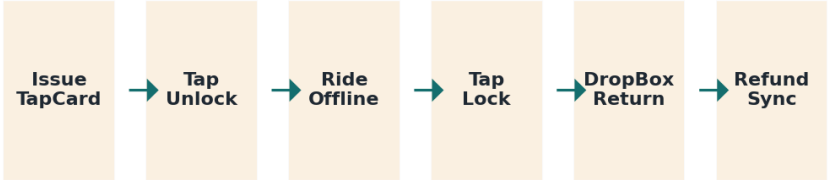
Tap 0.2s PN532 → BLE handover ID + odometer + battery. Ride offline logging Hall sensors + GPS. No data at rank, sync later at Checkers hub WiFi. AI wet brain detects slip: wheel vs GPS.

- PN532 0.2s
- BLE offline
- ESP32-S3



ISSUE → UNLOCK → RIDE OFFLINE → LOCK → HUB SYNC

NFC Edge-to-Edge Cycle



TapCard + DropBox — inclusive final lock

60% without NFC phones — deposit card makes Gordon's Bay cash-inclusive, DropBox closes the trust loop.

TapCard R12

- NTAG215 — last 10 trips
- Training flag + helmet size
- Balance lives on tablet, not card
- Deposit R50 / Buy R80

DropBox R650

- Steel Faraday cage grounded
- Slot 86×2mm labyrinth 90° bend
- IR beam wakes PN532 2s only inside
- Ferrite focus, RF 30% inward



No outward scanning: reader OFF until card breaks beam inside. Last TapReceiver writes RETURNED+LOCKED, drops to locked bin out of client hands. LED green = refund due.



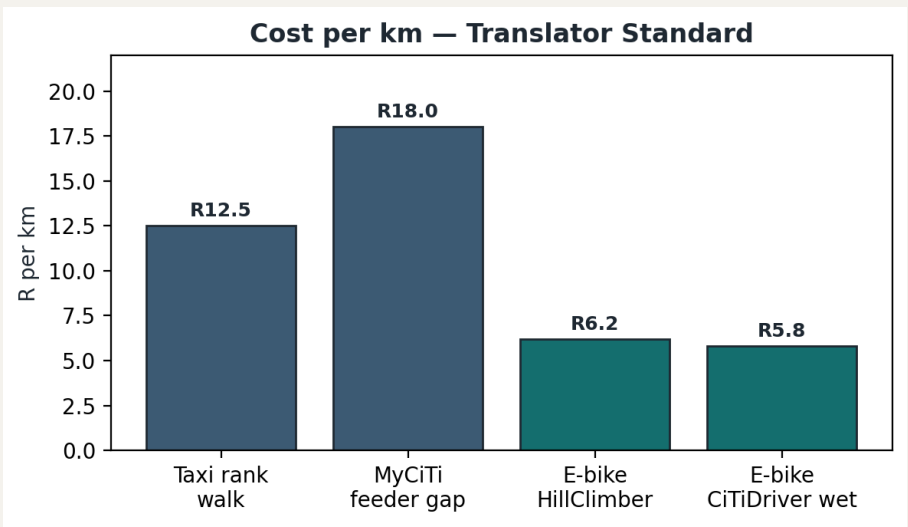
Blockbuster Box

Card received — R50 refund at controller desk

Business model – side roads funded by saved km

Cost per km translator funds infrastructure, not just rides. Fuel = 3× multiplier — e-bike wins, City saves, roads get patched.

COST PER KM – TRANSLATOR STANDARD



Formula

cost/km = base × method × fuel index
Taxi R12.5 walk, MyCiTi feeder R18.0 gap, HillClimber R6.2, CiTiDriver wet R5.8

Open standard

Auditable CSV



Savings engine

300 days airless zero puncture downtime, **65 rainy days** Wet Kit strap-on, motor derate saves battery. 0 roadside repairs.

0 punctures

300 days



Revenue

City: R6.2/feeder km per open km. **Sixty60: km share** per delivery. Deposit float **R50 × 200 cards = R10k** per hub.

MyCiTi

Sixty60



Jobs – support not own

Time Controller = TapCard agent + maintenance + safety trainer. 3 jobs per hub, 15 in pilot. Cash top-up, no bank needed.

3× per hub

Cash inclusive

Gordon's Bay pilot – 5 hubs at Checkers

Start where Checkers already has security, solar, CCTV — share roof, share cost. Support not own.

Checkers hub network

5 SITES READY

Central Checkers

Gordon's Bay main, Time Controller desk

Mountainside

15% grade start, HillClimber base

Harbour

Tourist + feeder, wet kit issue

Strand

Flat CiTiDriver feeder link

Somerset West rank

Taxi rank partnership, not competition

+ future infill

Open km data decides next patch

20

bikes

10 Hill + 10 CiTi

5

DropBoxes

steel Faraday

200

TapCards

R50 deposit

2

Cabinets

metal ventilated

90-day target — funded by saved km

5,000 km

feeder km audited open km standard

0

punctures — Tannus airless 300 days

100%

safety onboarding + QR brake log

50 m²

side-road patch funded by saved fuel

Data: anonymous trip ID, rounded 100m, export CSV for City, open source audited. No PII on card — balance on tablet only. ai.studio dashboard for City + Checkers + Sixty60.

✓ Solar LED

✓ CCTV share

✓ Jobs 3/hub

Ask — **partnership,** not procurement

What we need to launch Gordon's Bay in 30 days — no fleet ownership, just open km integration.



City of Cape Town

Approval for 5 Checkers hubs + feeder km rate per open km standard. Anonymous 100m CSV export.



Checkers

Share security / CCTV / solar at 5 sites, allow DropBox bolt (86mm slot). No new build.



Sixty60

4 Rider Cards, spare battery swap agreement, delivery km share for side-road fund.

OpenKM delivers:

- 10 HillClimber + 10 CiTiDriver
- 5 DropBoxes
- 200 TapCards
- 2 charging cabinets
- ai.studio dashboard
- Jobs training + open data audited